

* Number System Basics:-

e.g.

$$734 \rightarrow 700 + 30 + 4$$

$$= 7 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$$

Digits $\Rightarrow$ 0 to 9 10 digit

Base power 10

other number system:-

$\rightarrow$ Binary 2

$\rightarrow$ Octal - 8

$\rightarrow$ Hexadecimal - 16

Binary $\begin{cases} \rightarrow 0 \text{ and } 1 \text{ digit} \\ \rightarrow 2^1 \text{ power} \end{cases}$

$$\text{e.g. } (10100) = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 16 + 4$$

$$= 20$$

* Decimal to Binary:-

1. Repeatedly divide the number by 2, till you get 0.
2. Note down the remainders
3. Take the remainders in reverse.

2	34	
2	18	1
2	9	0
2	4	1
2	2	0
2	1	0
2	0	1

(100101)₂

* Adding 2 binary Numbers:-

		1	1				
7	1	0	1	1	0	22	5 = 1
	0	0	1	1	1	7	d = 10/2
	1	1	1	0	1	29	c = 1/2

* Why Binary:-

We humans use a decimal or base 10 numbering system, presumably because people have 10 digits.

Early computers were designed around the decimal numbering system. This approach made the creation of computer logic capabilities unnecessarily complex and did not make efficient use of resources. (for example, 10 vacuum tubes were needed to represent one decimal digit).

To deal with the basic electronic states of ON and OFF, von Neumann suggested using the binary system.

ON

1

and

OFF

0

* Bitwise operators:-

AND OR NOT XOR left right
shift shift
 $\&$ $|$ $\sim$ $\wedge$ $\ll$ $\gg$

Truth table:-

a	b	$a \& b$	$a b$	$a \wedge b$	$\sim a$	$\sim b$
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	1
1	1	1	1	0	0	0

↑
Addition without carry.

* Properties of Bitwise Operators:-

$$a \& 1 = \text{if } (a \& 1) = 1$$

n is odd

else

n is even.

$$1) a \& 0 = 0$$

$$2) 0 \& a = 0$$

$$3) a \cdot 0 = a$$

$$4) 0 \wedge 0 = 0$$

$$5) 0 \wedge 0 = 0$$

$$6) a \cdot 1 = \rightarrow \text{even} + 1$$

odd - 1

$$7) a \cdot 1 = \rightarrow \text{even} + 1$$

odd same

$$\begin{array}{r} 1 \ 1 \ 0 \ 1 \ 1 \\ 0 \ 0 \ 0 \ 1 \\ \hline 1 \ 0 \ 1 \ 1 \end{array}$$

$$8) a \neq b = b \neq a$$

$$9) a \cdot b = b \cdot a$$

$$10) a \wedge b = b \wedge a$$

Commutative property

Associate property

$$a \neq b \neq c = c \neq b \neq a$$

Example -:

$$\begin{aligned} & a \wedge b \wedge a \wedge d \wedge b \\ &= a \wedge a \wedge b \wedge b \wedge d \\ &= 0 \wedge 0 \wedge d \\ &= d \end{aligned}$$

* Single Number.:

arr[5] = 6 9 6 10 9
 ↙ ↘

Idea.:

Take XOR of all elements

TC: $O(N)$

SC: $O(1)$

* Left shift operator: ($\ll$)

8 bits 7 6 5 4 3 2 1 0

$a \ll 1$ 0 0 0 0 1 0 1 0 = 10
 — — — — — ↙ ↘ ↘ ↘

$a \ll 1$ 0 0 0 1 0 1 0 0 = 20
 — — — — —

$a \ll 2$ = 40 = 10×2^2

largest 8 bit number 255

Important Result.:

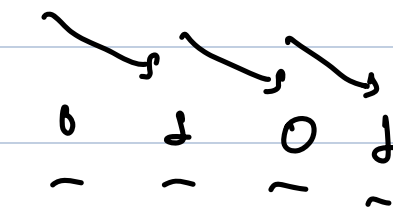
$$1 \ll N = 1 \times 2^N$$

$$= 2^N \quad \leftarrow \quad O(1) \text{ time}$$

A Right shift operator: ($\gg$)

$$a = 10$$

$$\begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{0} \end{array} = 10$$

$$a \gg 1 \quad \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} \end{array} = 10/2$$


$$a \gg 2 = 10/4$$

Generalisation:

$$a \gg i = \frac{a}{2^i}$$